

Pysteps in Operations

Some background to nowcasting and how it can be used



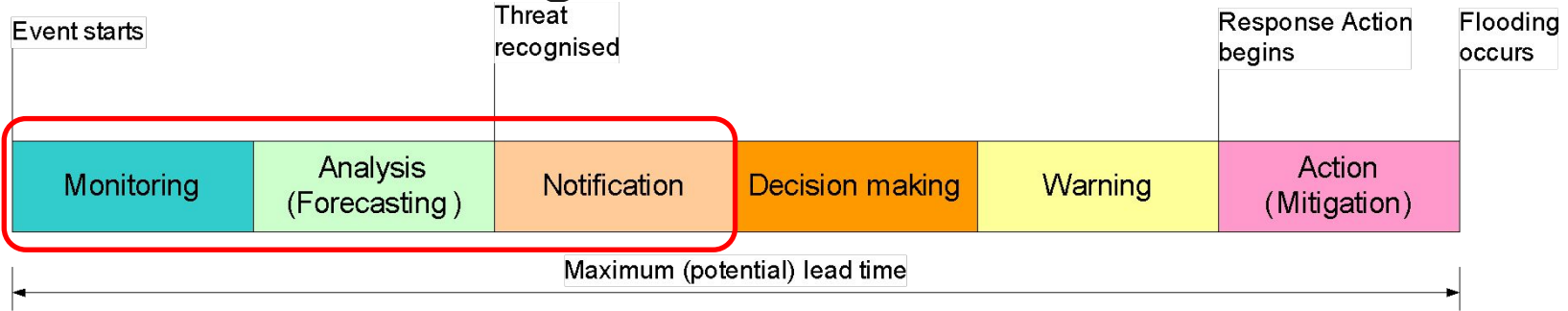
Session overview

Time	Content	Speaker	Duration
09:00	Welcome / Course overview	Ruben	20'
09:20	Lecture: "Pysteps in operations"	Ruben, Lesley	40'
10:00	Coffee break		30'
10:30	Hands-on session "Starting with pysteps"	Jenna, Miguel, Mats	120'
12:30	Lunch break		60'
13:30	Hands-on session "Blending with NWP"	Mats, Lesley, Ruben	120'
15:30	Coffee break		30'
16:00	Lecture: "AI for nowcasting"	Lesley, Jenna, Ruben	45'
16:45	Discussion	All	15'
17:00	End of the short course		

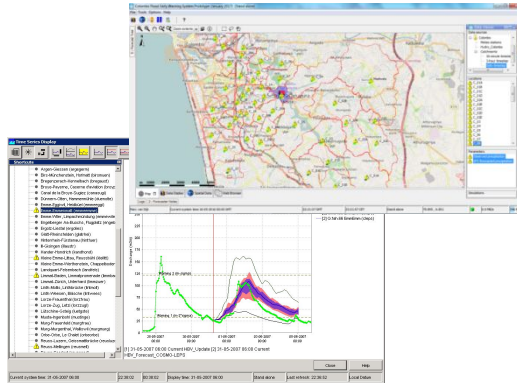


Why we need improved rainfall forecasts for short lead times

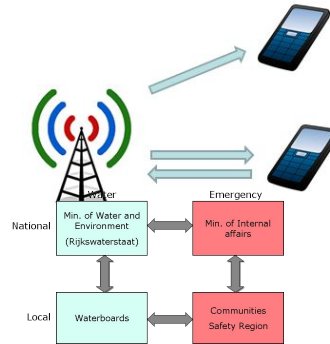
Forecasts, warning and action



Forecast



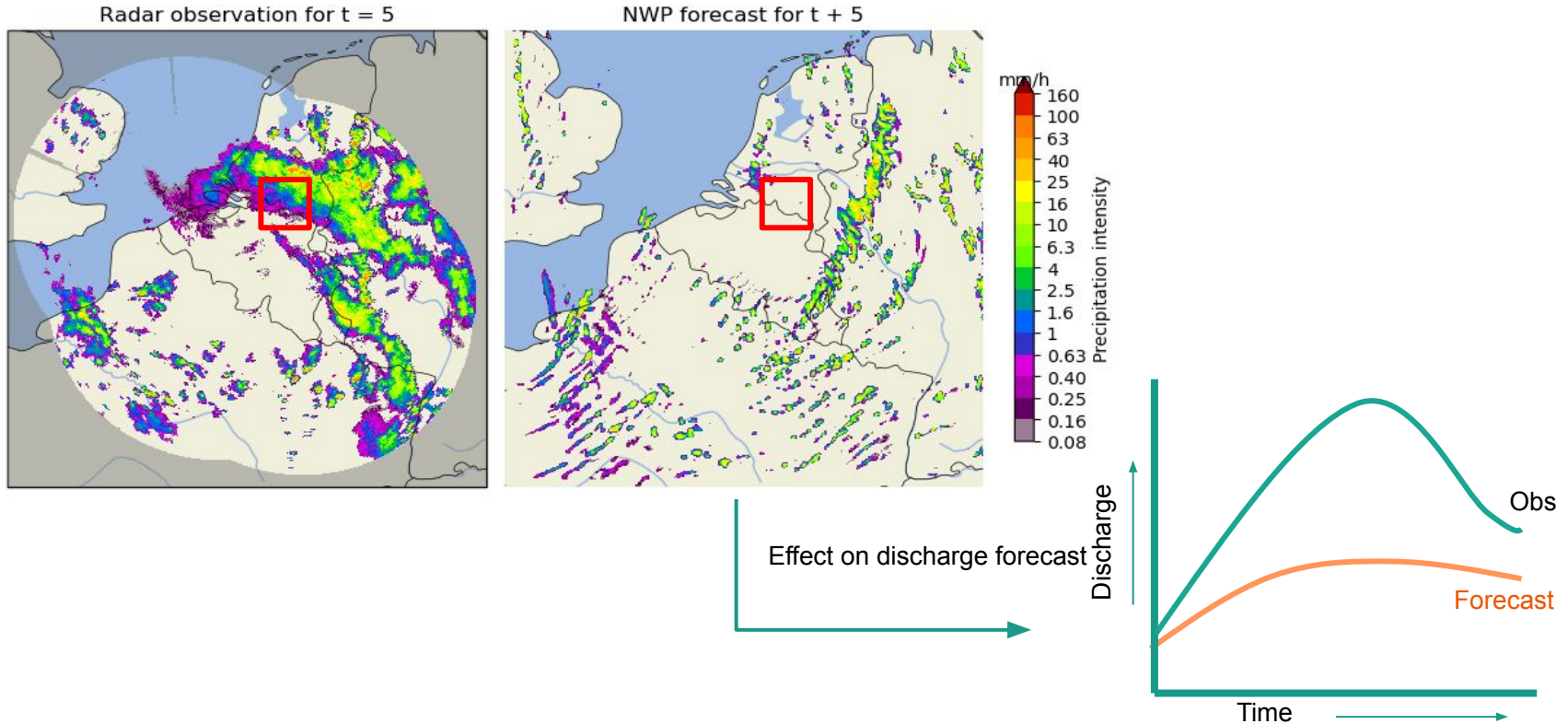
Warning



Action



The limitation of NWP models on short lead times for flood early warning



What is nowcasting and why should we use it?

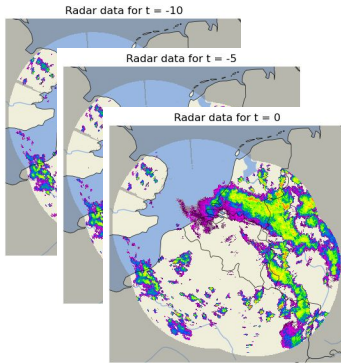


How do we make a nowcast?

Extrapolation

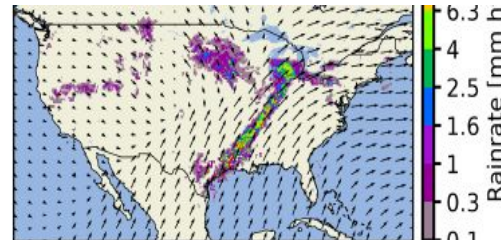
Fast!

But no physics involved..



From most recent weather
radar observations

Determine
movement of
rainfall fields



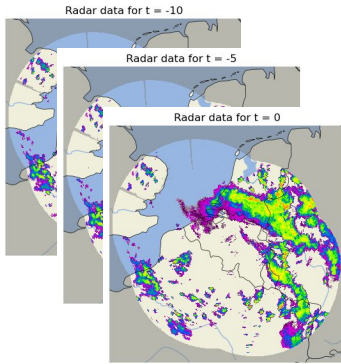
Source: Pulkkinen et al., Geosci. Model. Dev., 2019

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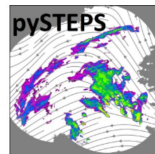
From most recent weather
radar observations

Determine
movement of
rainfall fields

Rainfall field
evolution



Source: Pulkkinen et al., Geosci. Model. Dev., 2019

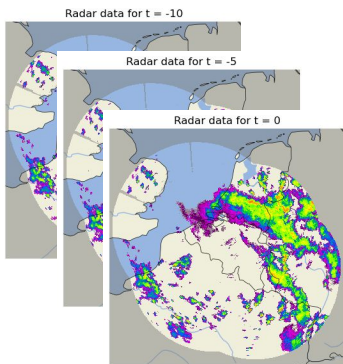


How do we make a nowcast?

Extrapolation

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But no physics involved..



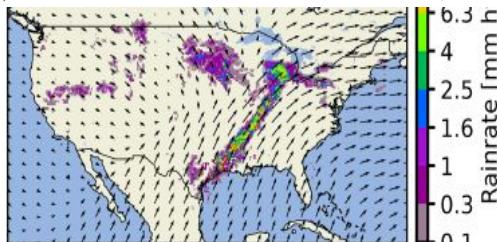
From most recent weather
radar observations

Determine
movement of
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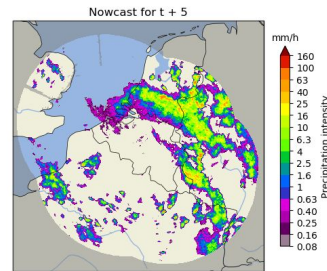
Rainfall field
evolution

Add
stochastic
noise

Extrapolate
to future



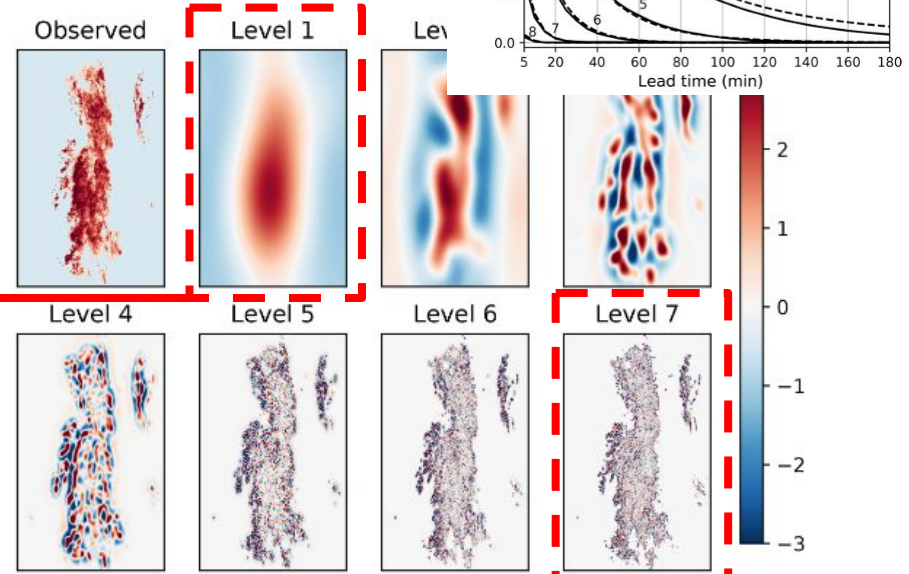
Source: Pulkkinen et al., Geosci. Model. Dev., 2019



To rainfall forecast up to max.
6 hr ahead

The STEPS algorithm in pysteps

- Past research has shown that the lifetime of rainfall fields relates to its spatial scale (e.g., Venugopal et al., 1999, j. Geophys. Res.-Atmos.)
- STEPS takes this into account by decomposing the rainfall fields in spatial cascades with a fast Fourier transform.

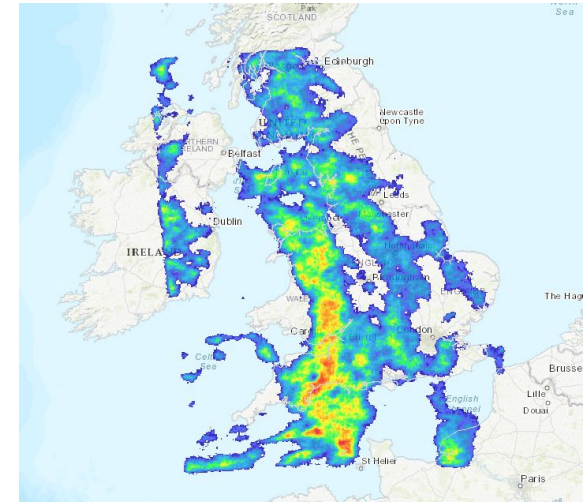
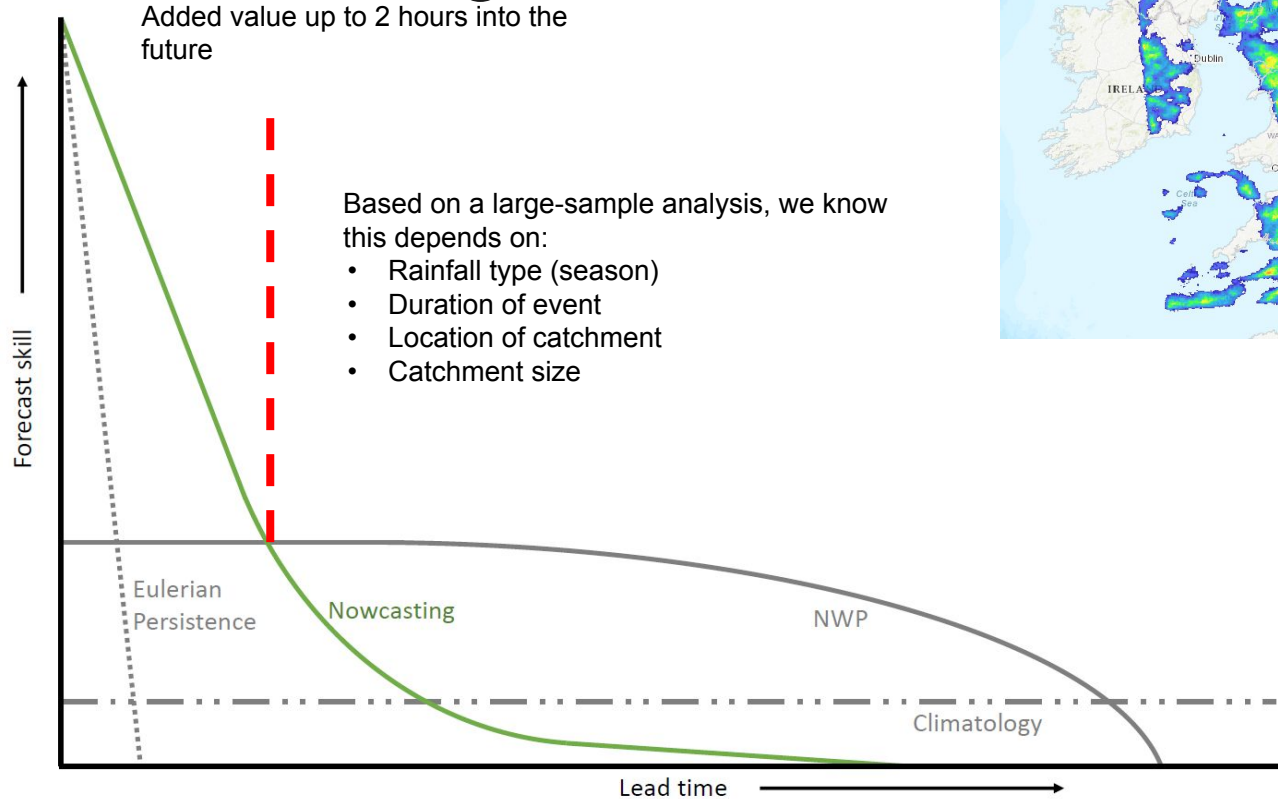


Largest spatial scale: **long lifetime**, advects (moves) without much growth and dissipation

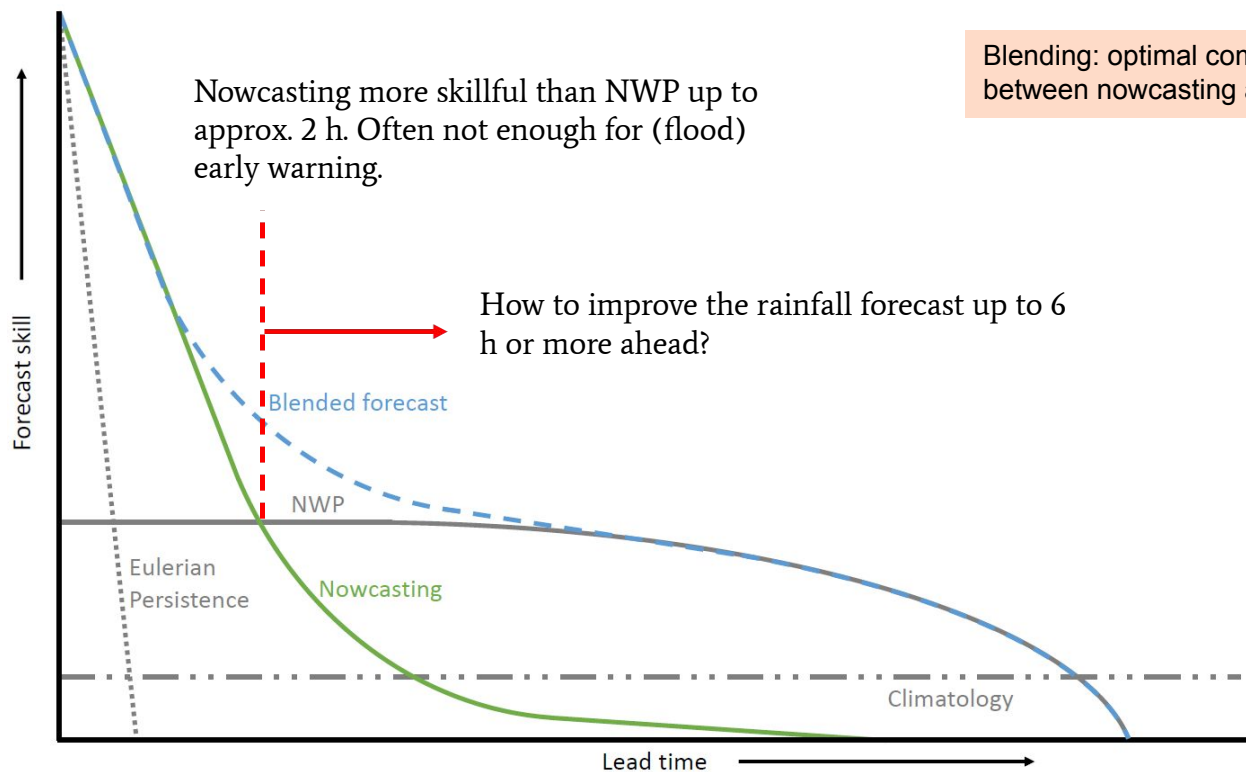
Smallest spatial scale: **short lifetime**, advects (moves) and changes with a combination of **autoregression** and **stochastic perturbations (to model growth and dissipation)** using spatially correlated Gaussian noise.

Source: Pulkkinen et al., 2019, GMD

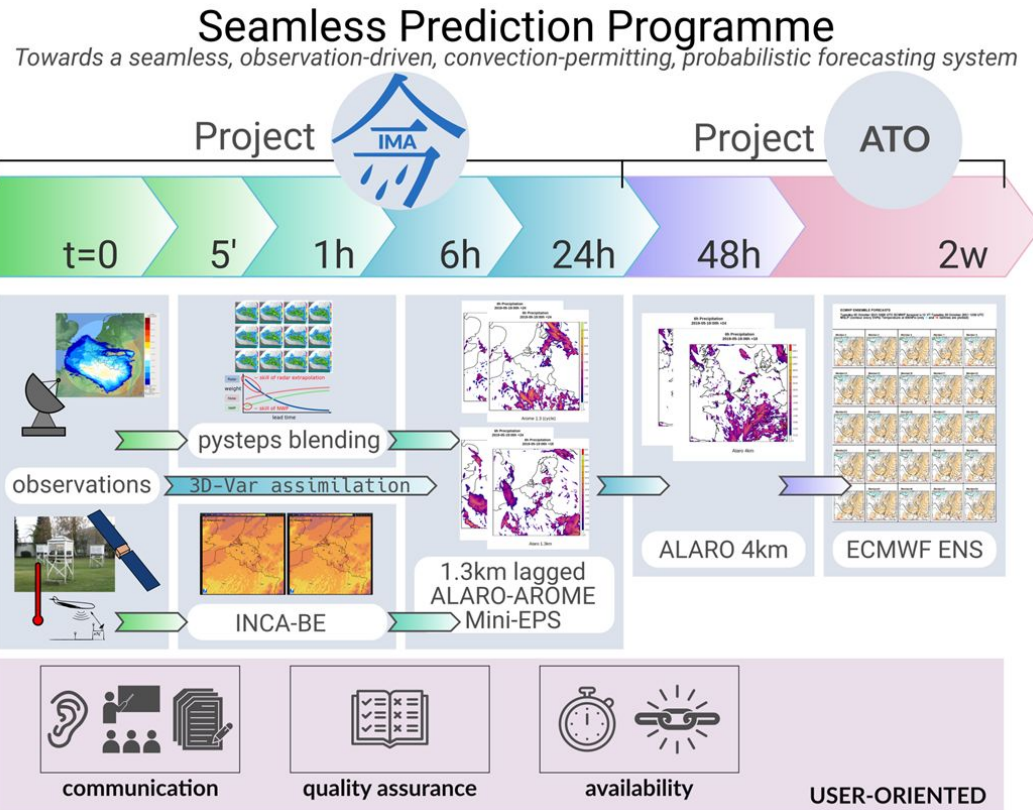
The skill of nowcasting



Can we even go a step further?

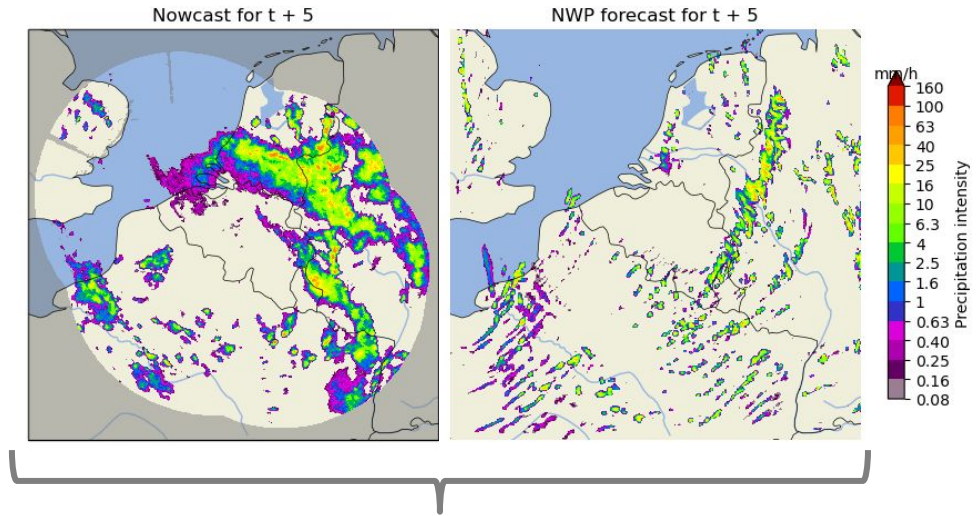
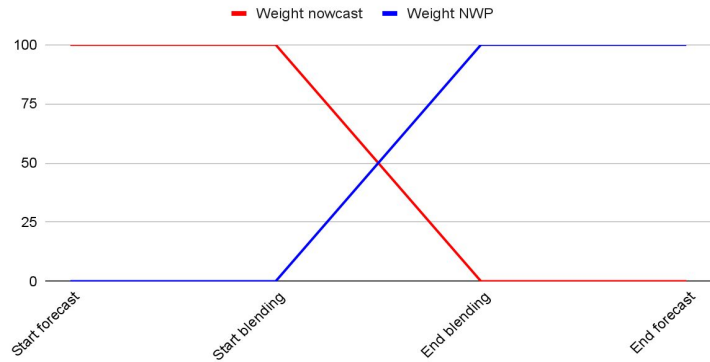


Seamless forecasting



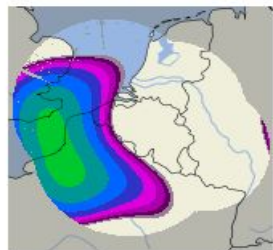
Simple linear blending?

Linear blending weights

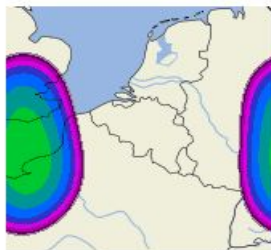


How to combine
a case like this?

NWC cascade level 0
2022-05-20 08:00 UTC + 5 min



NWP cascade level 0
2022-05-20 00:00 UTC + 485 min



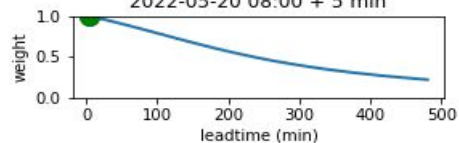
Noise cascade level 0
2022-05-20 08:00 UTC + 5 min



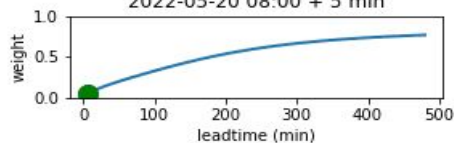
Blended cascade level 0
2022-05-20 08:00 UTC + 5 min



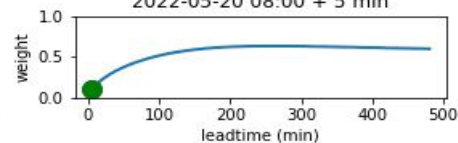
NWC weight cascade level 0
2022-05-20 08:00 + 5 min



NWP weight cascade level 0
2022-05-20 08:00 + 5 min



Noise weight cascade level 0
2022-05-20 08:00 + 5 min



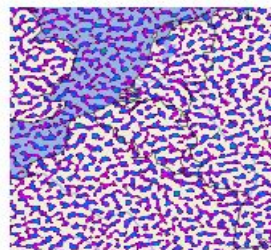
NWC cascade level 4
2022-05-20 08:00 UTC + 5 min



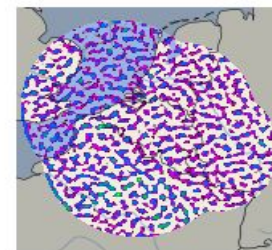
NWP cascade level 4
2022-05-20 00:00 UTC + 485 min



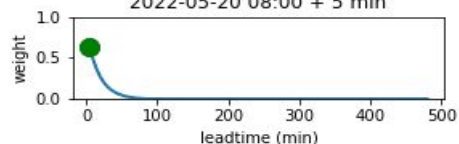
Noise cascade level 4
2022-05-20 08:00 UTC + 5 min



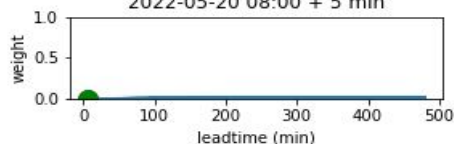
Blended cascade level 4
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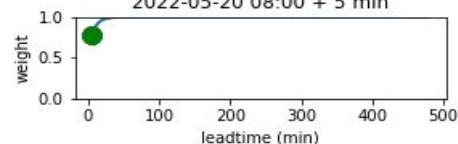
NWC weight cascade level 4
2022-05-20 08:00 + 5 min



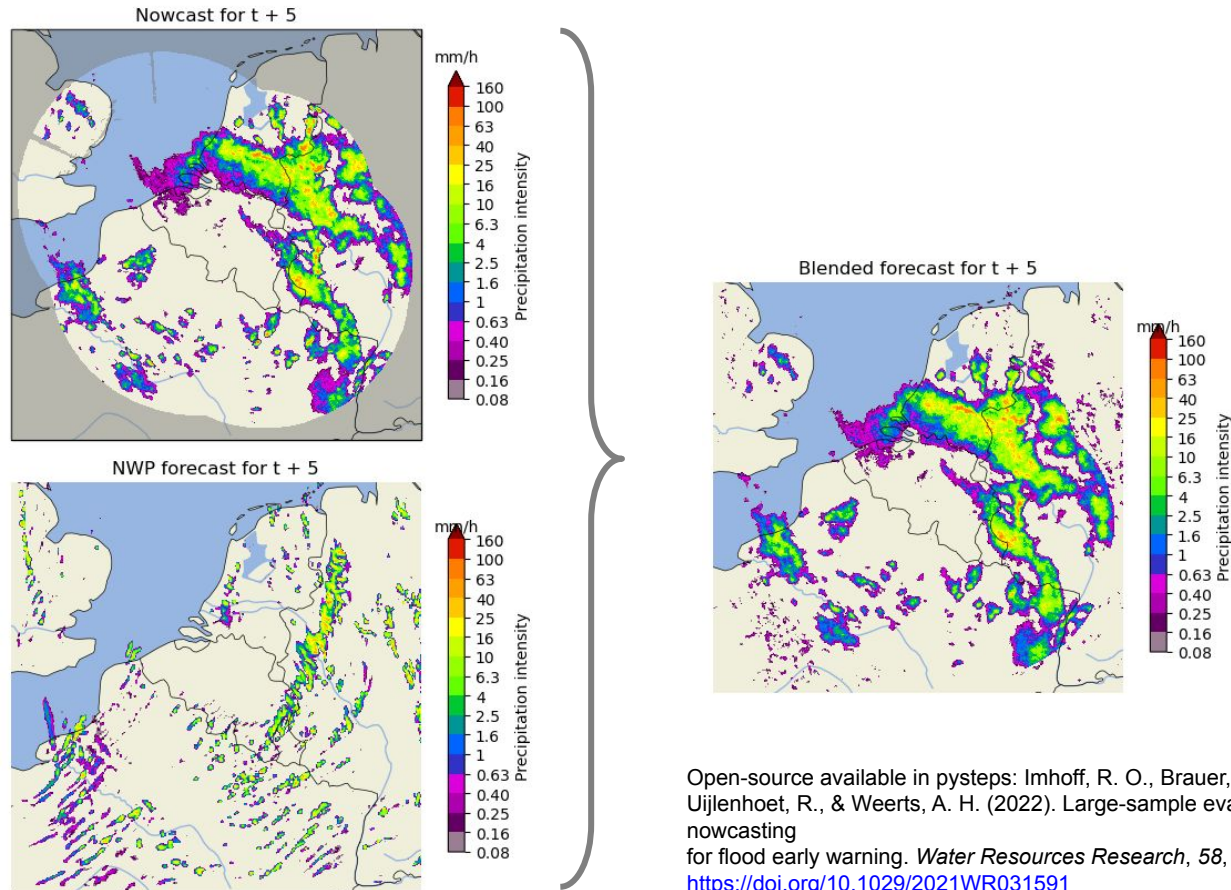
NWP weight cascade level 4
2022-05-20 08:00 + 5 min



Noise weight cascade level 4
2022-05-20 08:00 + 5 min

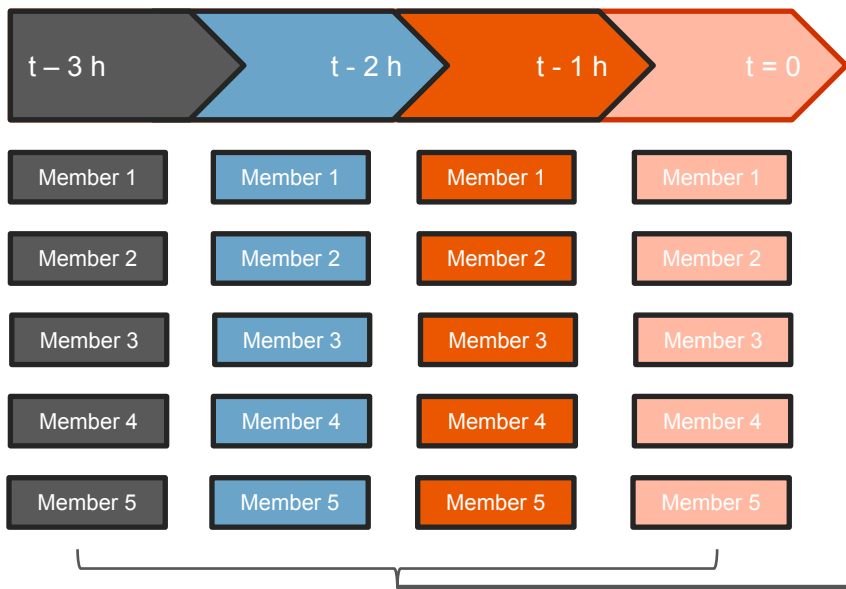


An advanced blending method



Open-source available in pysteps: Imhoff, R. O., Brauer, C. C., van Heeringen, K. J., Uijlenhoet, R., & Weerts, A. H. (2022). Large-sample evaluation of radar rainfall nowcasting for flood early warning. *Water Resources Research*, 58, e2021WR031591. <https://doi.org/10.1029/2021WR031591>

Going seamless: Operationalization of pysteps at KNMI

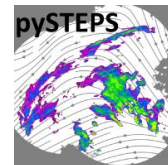
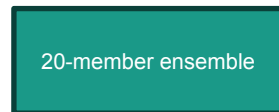


Deterministic nowcast:

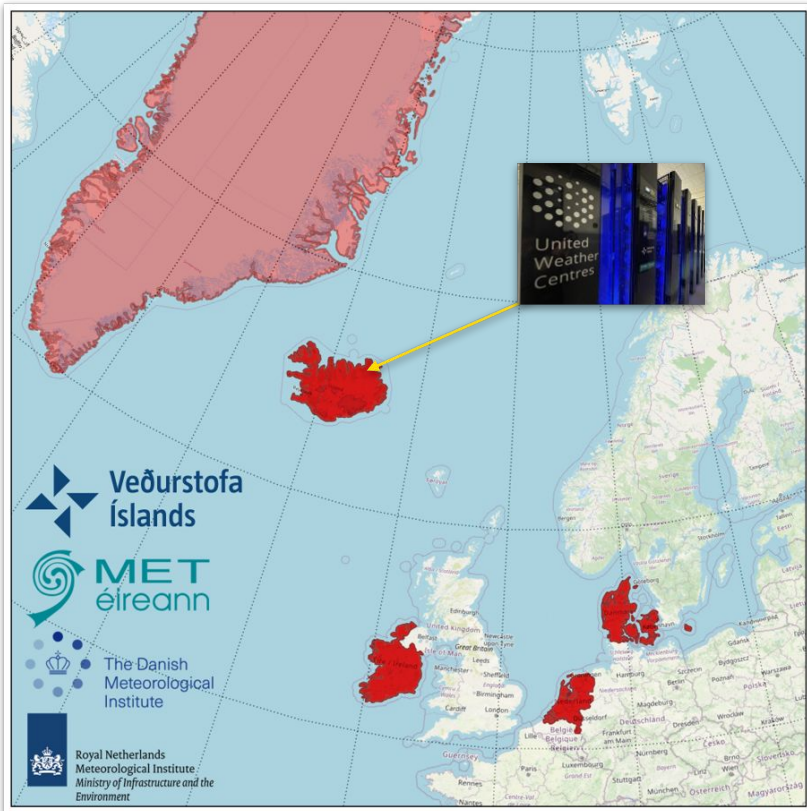
- Issued every 5 min.
- Up to 2 hours ahead.
- Only nowcast for visualization in app, etc.

Seamless forecast (blending):

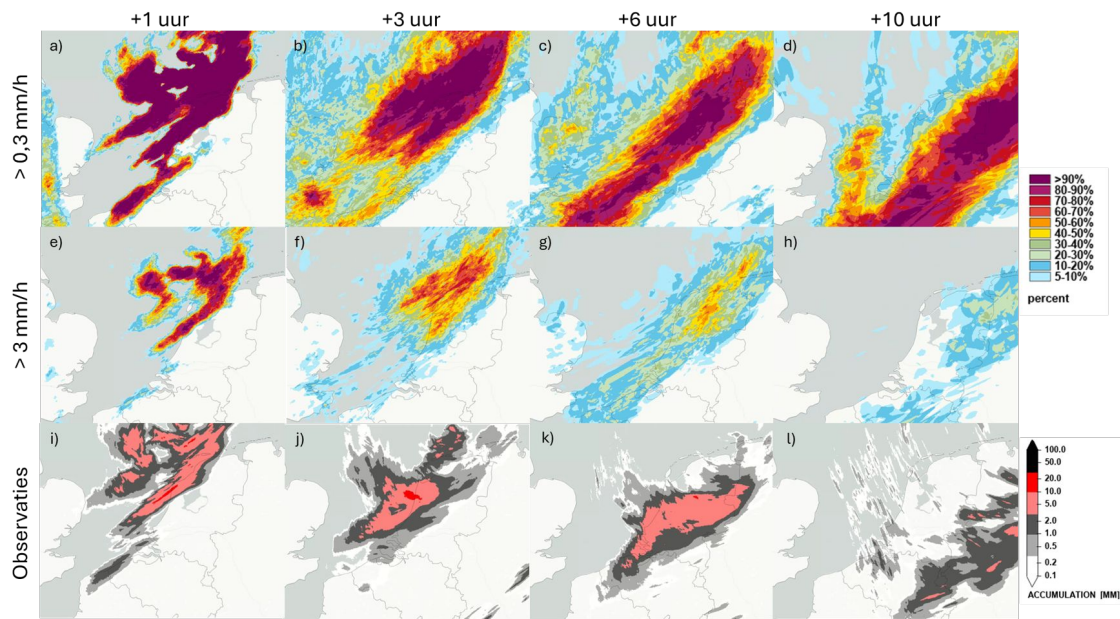
- Seamless ensemble forecast
- Issued every 5 min.
- Up to 6 hours ahead then 100% NWP.
- 20 Ensemble members from lagged NWP forecast.



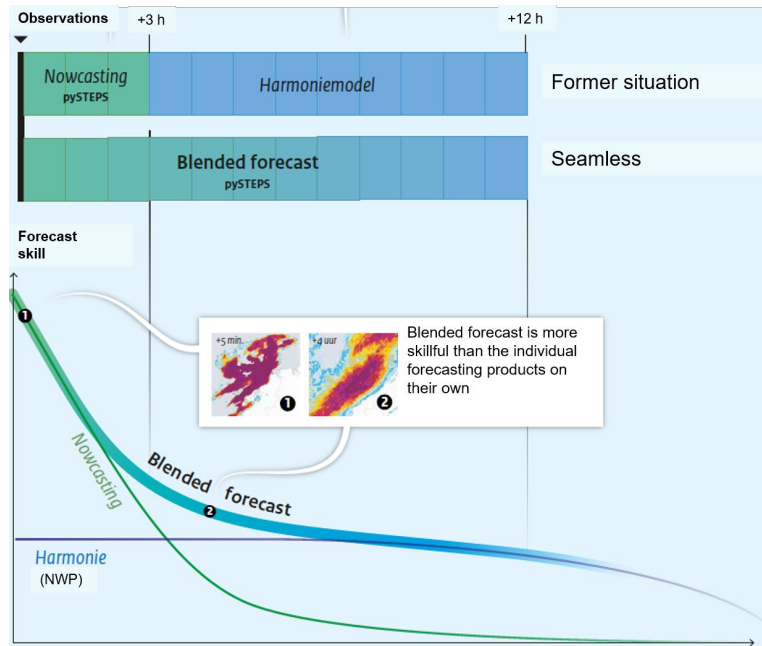
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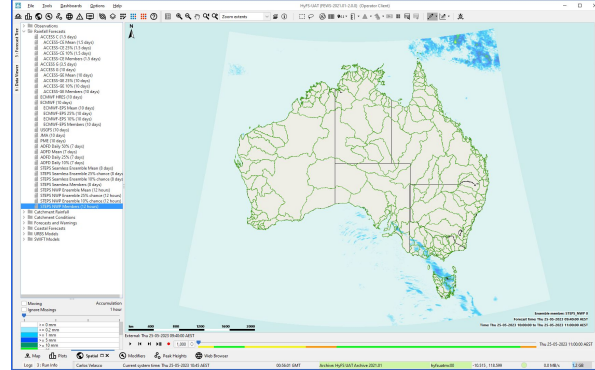
Going seamless: Operationalization of pysteps at KNMI



What's next? → NWP at 15-min time step, so better matching with nowcast in blending!

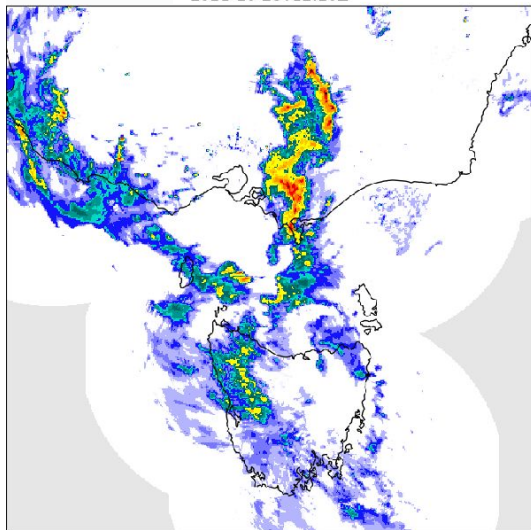


STEPS-NWP at Bureau of Meteorology, Australia



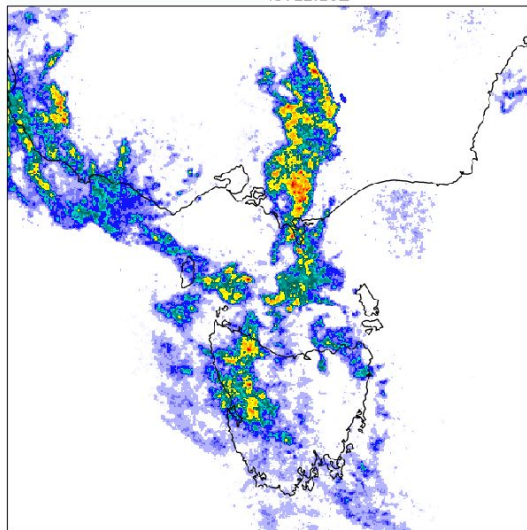
RADAR

Red radar rainfall rate (10min)
Vic-Tas domain
2021-10-28T12:10Z



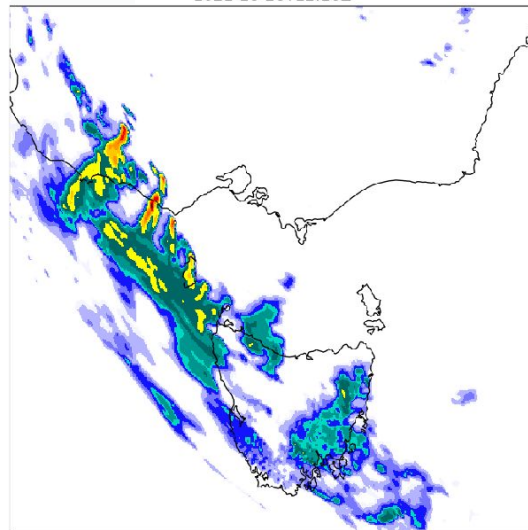
STEPS-NWP

STEPS-NWP rainfall rate
2021-10-28T12:00
adtime -10 mins
2021-10-28T12:10Z



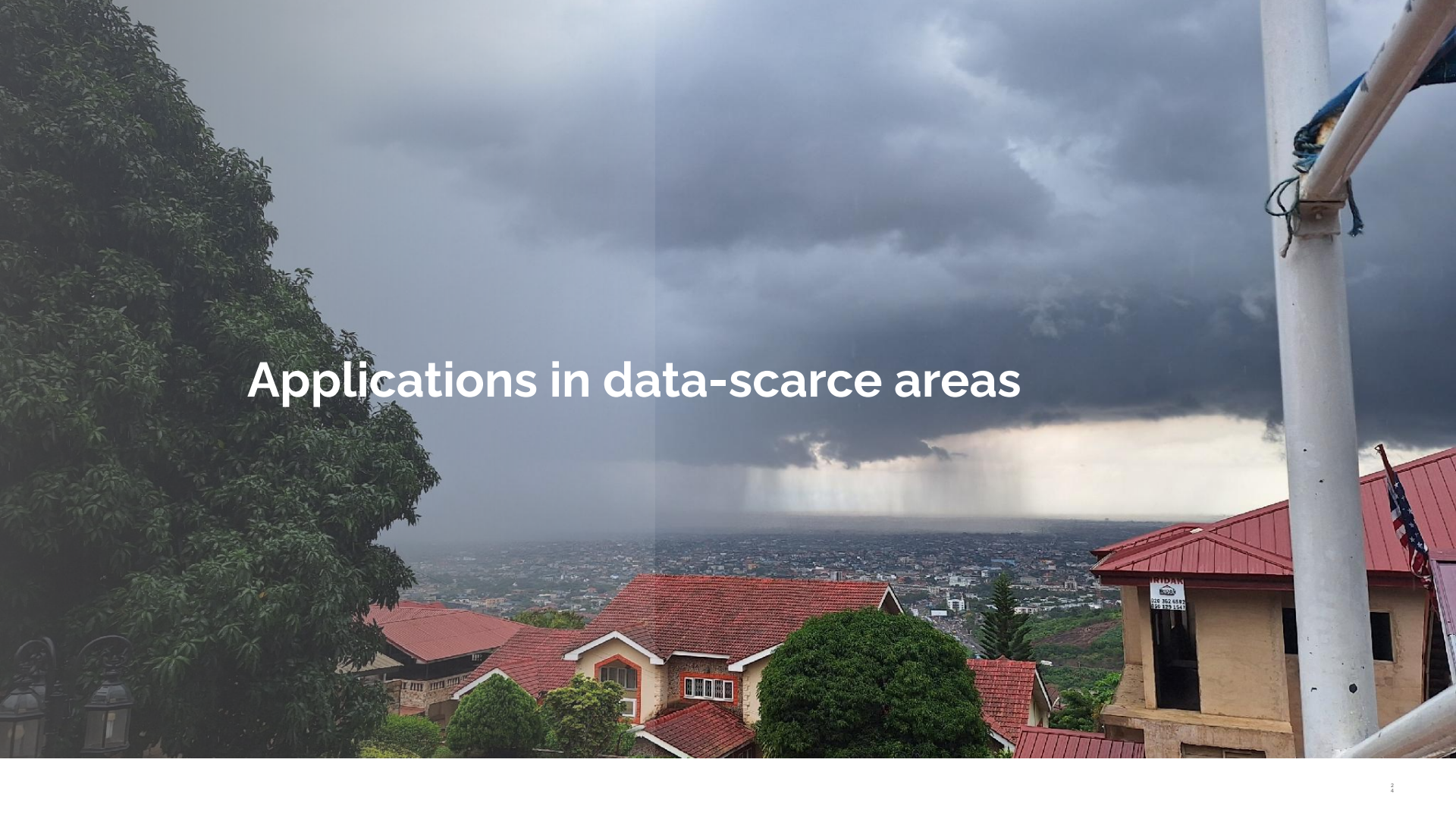
NWP

Blended G3CE3 rainfall rate
Run time : 2021-10-28T06:00
Member - 03
2021-10-28T12:10Z



Going seamless: Operationalization of pysteps at RMI

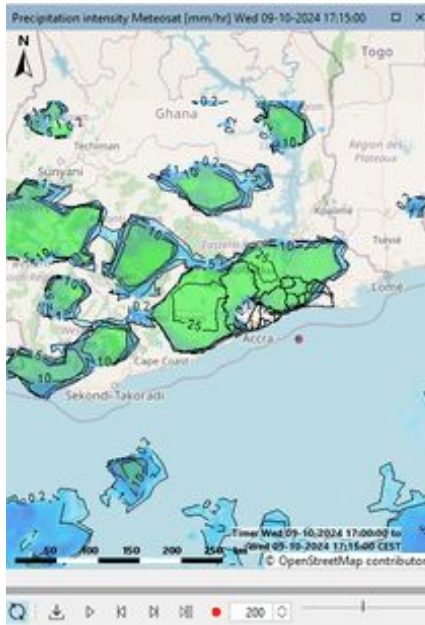
Other Applications?

A dramatic landscape photograph showing a cityscape under a heavy, dark, stormy sky. In the foreground, there are houses with red-tiled roofs and lush green trees. A white utility pole is visible on the right side. The text "Applications in data-scarce areas" is overlaid in the center of the image.

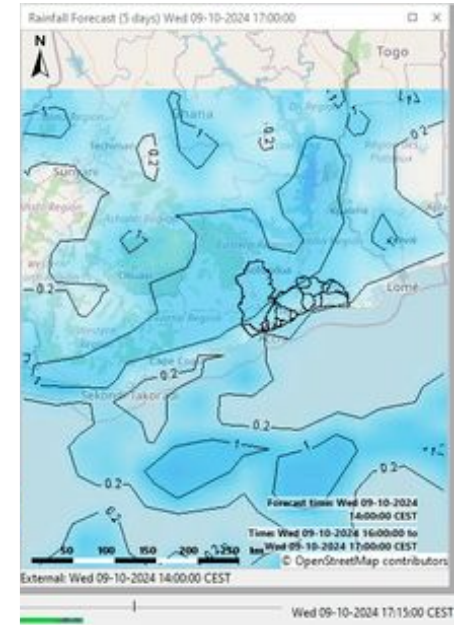
Applications in data-scarce areas

Satellite-based rainfall nowcasting as alternative

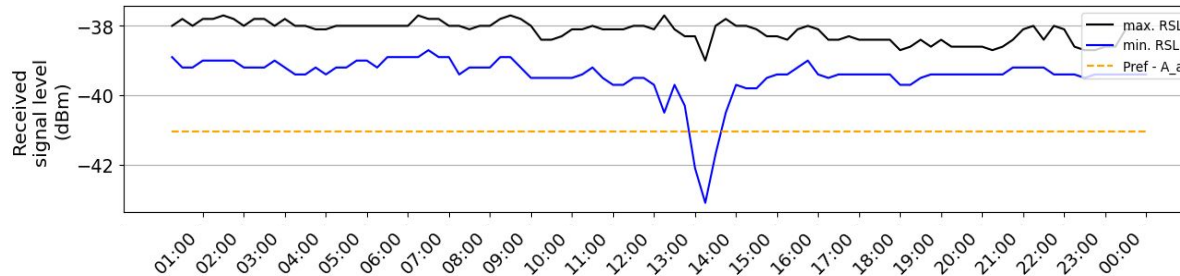
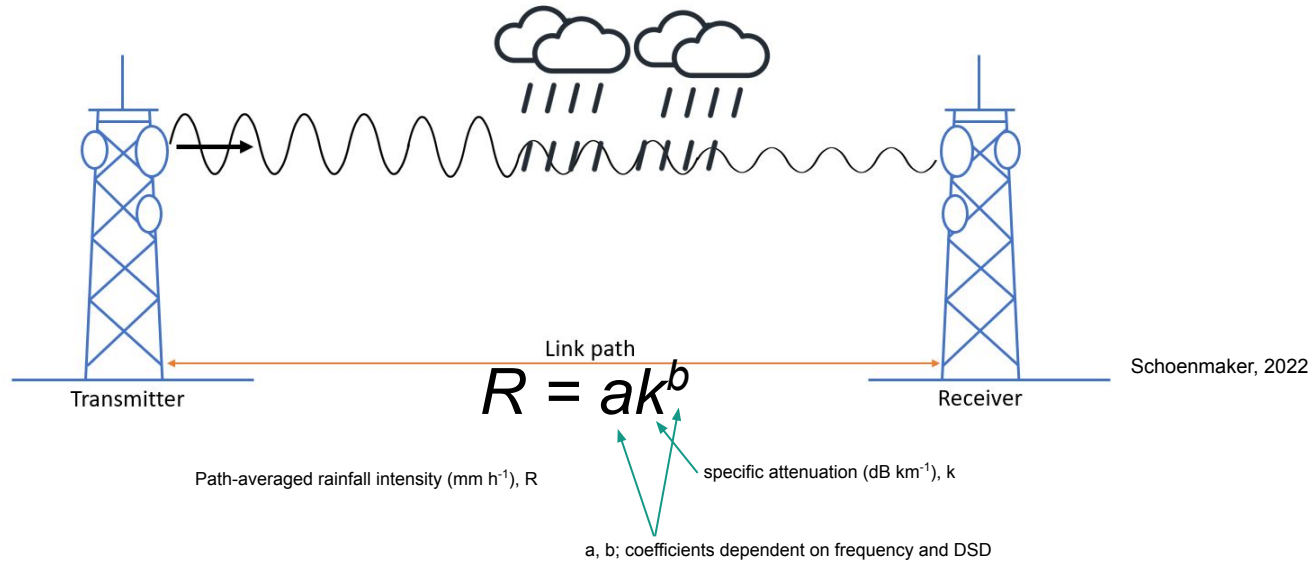
Meteosat observations STEPS



ge forecast



Rainfall forecasting with alternative sensors

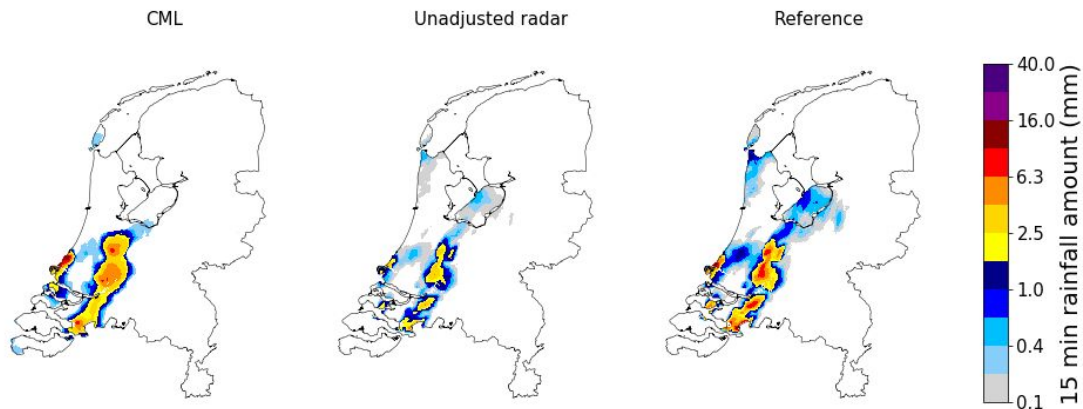


Walraven et al., 2026,
WRR (in review)

Rainfall forecasting with alternative sensors

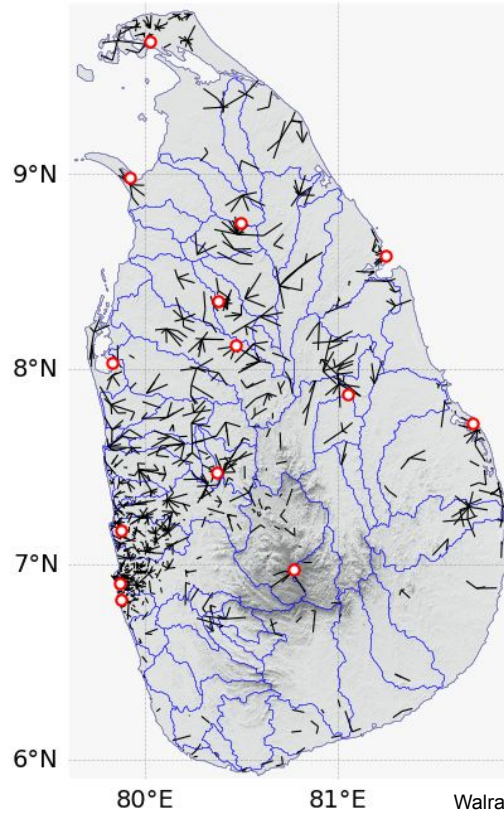
Application for the Netherlands

2011-09-10 20:00 UTC + 0 min.



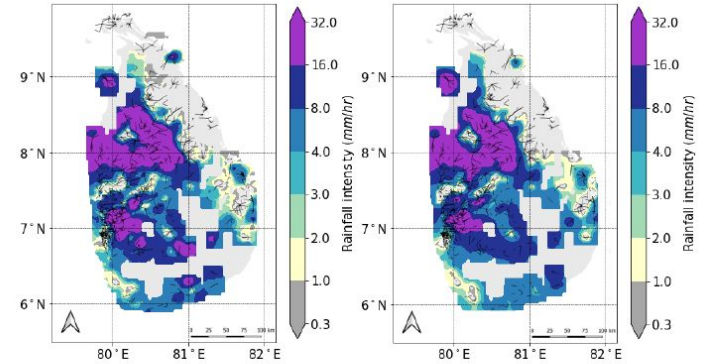
Courtesy of Bas Walraven

Rainfall forecasting with alternative sensors



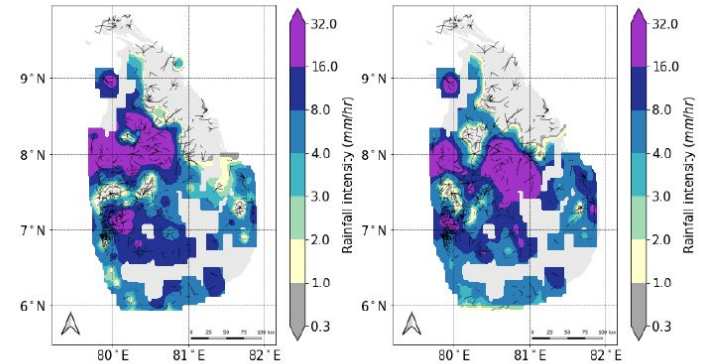
Walraven et al., 2026,
WRR (in review)

Application for Sri Lanka



(a) QPE at 6:00 pm

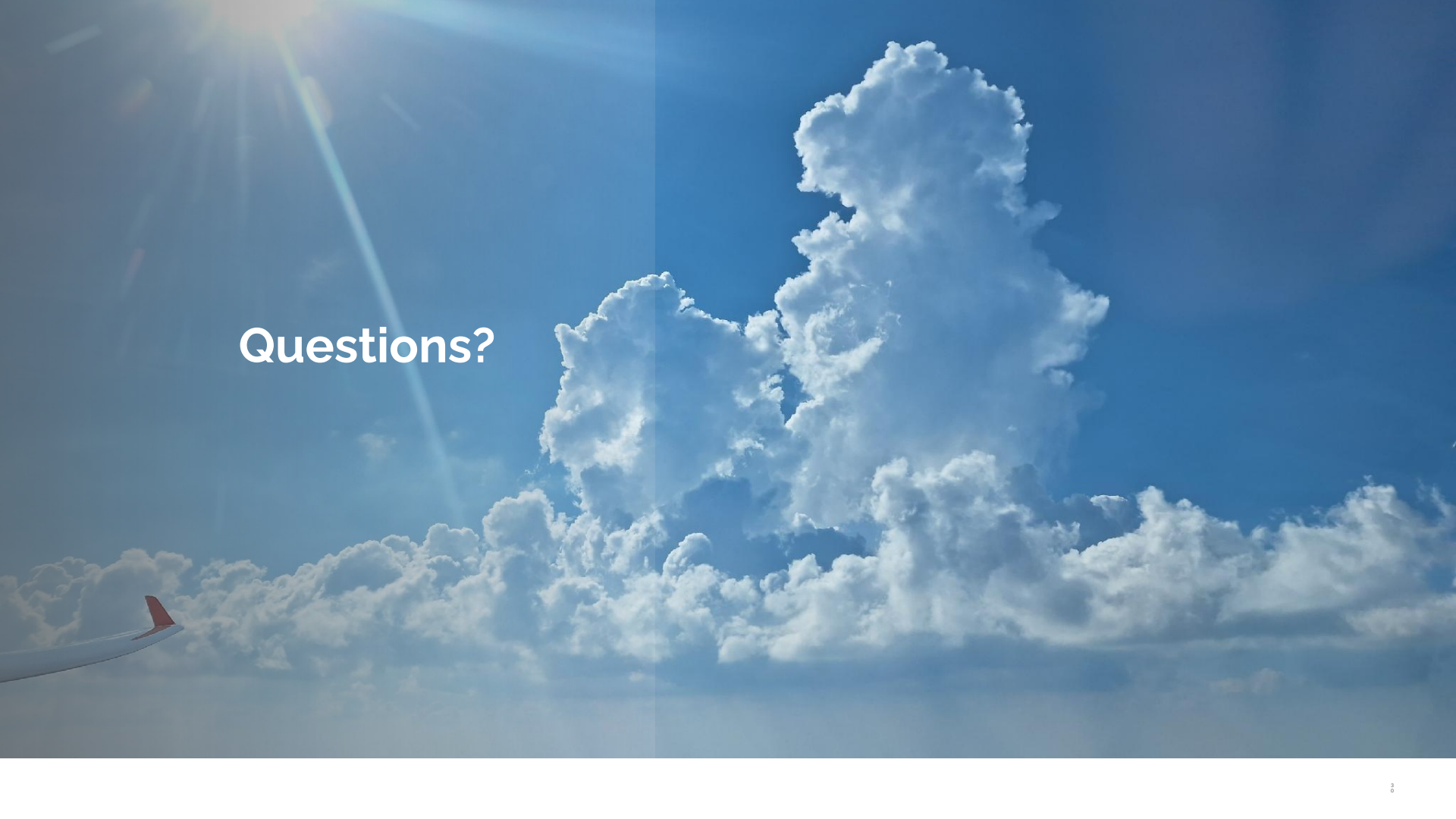
(b) Pysteps-D: 15 min lead time



(c) Pysteps-D: 30 min lead time

(d) Pysteps-D: 60 min lead time

Schoenmaker, 2022



Questions?



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